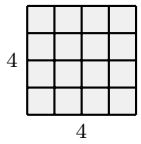
# Lesson 1: Understanding Powers and the Exponent Laws

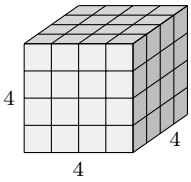
*Base, exponent, coefficient — then the five exponent laws rebuilt from repeated multiplication.*

## 1. What Does an Exponent Mean?

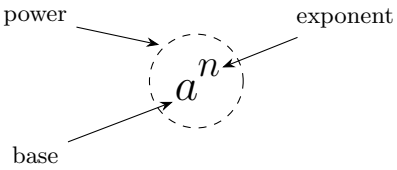
An exponent is shorthand for repeated multiplication of the *same* factor: $6\times6\times6\times6=6^4$, and likewise $p\times p\times p\times p\times p=p^5$. Writing a repeated multiplication this compact way is called **exponential form**. A **power** is any quantity written that way — the *whole* expression $a^n$, not just the little number. Its **base** is the factor being repeated, and its **exponent** tells how many times the base is used as a factor.




$4\times4=4^2=16$ *squares*
*"four **squared**"*




$4\times4\times4=4^3=64$ *cubes*
*"four **cubed**"*




*The whole expression $a^n$ is the power.*

A number multiplying a power is its **coefficient**: in $9m^4$ the coefficient is 9, the base is $m$, and the exponent is 4. If no number is written in front of a power, the coefficient is understood to be 1.

**Example 1 — Naming the coefficient, base and exponent**

For each expression below, state the coefficient (if any is written), the base, and the exponent.

**(a)** $9^4$ **(b)** $t^{12}$ **(c)** $5^y$ **(d)** $(-w)^7$

**(e)** $\left(\frac{2}{5}\right)^8$ **(f)** $6x^5$ **(g)** $-4z^3$ **(h)** $\frac{3y^9}{7}$

## 2. Coefficients vs. Powers of a Product

Grouping matters. In $5m^3$ only $m$ is repeated three times; the 5 sits out front as a coefficient. In $(5m)^3$ the *entire* quantity $5m$ is the base, repeated three times:

$$(5m)^3=(5m)\times(5m)\times(5m)=(5\times5\times5)\times(m\times m\times m)=125m^3.$$

Parentheses act like a fence: only what is *inside* gets repeated by the exponent, and anything outside is unaffected.

**Example 2 — Where the fence sits**

Write each of the following as a repeated multiplication.

**(a)** $5m^3n$ **(b)** $5mn^3$

**(c)** $5(mn)^3$ **(d)** $(5mn)^3$

## 3. Evaluating Powers, Including Negative Bases

The fence also settles a subtlety with negative numbers. In $-7^2$ no parentheses enclose the $-7$, so the fence holds only the 7 and the minus sign sits outside the power: $-7^2=-\left(7^2\right)=-49$. In $(-7)^2$ the fence holds $-7$ itself, so $-7$ is the base that gets multiplied by itself: $(-7)^2=(-7)\times(-7)=49$. When the exponent is *odd* the distinction disappears, since a lone negative sign out front behaves the same as an odd number of negative factors multiplied together.

**Example 3 — $-7^2$ is not $(-7)^2$**

Evaluate each of the following without a calculator.

**(a)** $-7^2$ **(b)** $(-7)^2$ **(c)** $-7^3$ **(d)** $(-7)^3$

### 4. The Zero Exponent

Dropping the exponent of 6 by one divides the value by 6:

$$6^4 = 1296, \qquad 6^3 = 216, \qquad 6^2 = 36, \qquad 6^1 = 6, \qquad 6^0 = ?$$

so $6^0 = 6 \div 6 = 1$. The pattern holds for any nonzero base, giving the **Zero Exponent Law** $a^0 = 1$ $(a \neq 0)$. Watch the fence: in $-a^0$ the exponent applies only to $a$, so $-a^0 = -(a^0) = -1$.

**Example 4 — Zero exponents, inside and outside the fence**

Evaluate each of the following.

**(a)** $11^0$ **(b)** $(-8)^0$ **(c)** $-8^0$ **(d)** $3\left(5^2\right)^0$

### 5. The Five Exponent Laws

Each law comes from expanding the powers into repeated multiplication. The first column shows the expansion with a numerical base (6); the same thing happens with variable bases ($c$ and $d$).

| Expanded (numerical base) | Variable base | Exponent Law |
|---|---|---|
| $6^3 \times 6^2 = (6\cdot6\cdot6)(6\cdot6) = 6^5$ | $c^3 \times c^2 = c^5$ | **Product Law:** $a^m \cdot a^n = a^{m+n}$ |
| $6^5 \div 6^2 = \dfrac{6\cdot6\cdot6\cdot\cancel{6}\cdot\cancel{6}}{\cancel{6}\cdot\cancel{6}} = 6^3$ | $c^5 \div c^2 = c^3$ | **Quotient Law:** $a^m \div a^n = \dfrac{a^m}{a^n} = a^{m-n}$, $(a \neq 0)$ |
| $(6\cdot5)^3 = (6\cdot5)(6\cdot5)(6\cdot5) = 6^3\cdot5^3$ | $(cd)^3 = c^3d^3$ | **Power of a Product Law:** $(ab)^m = a^m b^m$ |
| $\left(\dfrac{6}{5}\right)^3 = \dfrac{6}{5}\cdot\dfrac{6}{5}\cdot\dfrac{6}{5} = \dfrac{6^3}{5^3}$ | $\left(\dfrac{c}{d}\right)^3 = \dfrac{c^3}{d^3}$ | **Power of a Quotient Law:** $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$, $(b \neq 0)$ |
| $\left(6^3\right)^2 = (6\cdot6\cdot6)(6\cdot6\cdot6) = 6^6$ | $\left(c^3\right)^2 = c^6$ | **Power of a Power Law:** $(a^m)^n = a^{mn}$ |

**Example 5 — Simplify with a law, then evaluate**

Use the exponent laws to simplify, then evaluate.

**(a)** $4^3 \cdot 4^2$ **(b)** $\dfrac{(-3)^6}{(-3)^2}$ **(c)** $\left(7^2\right)^2$

**Example 6 — All five laws with variable bases**

Use the exponent laws to simplify.

**(a)** $k^5 \cdot k^3$ **(b)** $m^2 \cdot m^0 \cdot m^6$ **(c)** $\dfrac{n^9}{n^4}$ **(d)** $\left(p^4\right)^3$

**(e)** $(2r)^4$ **(f)** $\left(\dfrac{s}{t}\right)^5$ **(g)** $(-5uv)^3$ **(h)** $\left(\dfrac{w}{4}\right)^2$

## Lesson 2: Simplifying Expressions with Several Exponent Laws

*Combining the product, quotient and power laws in one chain — plus negative bases and variable exponents.*

### 1. Recap: Expanding Into Factors

Lesson 1, Section 5 built each law by expanding into factors. The same check works when there are coefficients — write each term in **expanded factor form**:

**Multiplying**

$$\begin{aligned} & 3x^2 \times 2x^4 \\ &= (3 \cdot x \cdot x) \times (2 \cdot x \cdot x \cdot x \cdot x) \\ &= (3 \times 2) \cdot (x \cdot x \cdot x \cdot x \cdot x \cdot x) \\ &= 6x^6 \end{aligned}$$

**Dividing**

$$\begin{aligned} & 12a^5 \div 3a^2 \\ &= \frac{12 \cdot a \cdot a \cdot a \cdot a \cdot a}{3 \cdot a \cdot a} \\ &= \frac{12}{3} \cdot a \cdot a \cdot a \\ &= 4a^3 \end{aligned}$$

Coefficients combine like ordinary numbers; the variable factors stack up (Product Law) or cancel (Quotient Law).

**Example 1 — Three students, three answers**

Three students are asked to simplify $5x^2 \times 4x^4$. Their answers are shown below.

Devon $\Rightarrow 20x^8$ Farah $\Rightarrow 9x^6$ Priya $\Rightarrow 20x^6$

Use expanded factor form to explain which student's answer is correct, and describe the mistake made by each of the other two.

**Example 2 — Justifying the quotient law with factors**

Use expanded factor form to explain why $10a^8 \div 5a^3 = 2a^5$.

### 2. A Strategy for Combining Several Laws

With several laws at once, work in this order:

- **Step 1:** Raise each *inner* bracket to its power (Power of a Product / Power of a Quotient / Power of a Power) — every factor inside, coefficient included. (A bracket around the whole expression waits for Step 4.)
- **Step 2:** Multiply or divide the coefficients like ordinary numbers.
- **Step 3:** Add or subtract exponents, one base at a time (Product and Quotient Laws).
- **Step 4:** If the whole expression sits inside an outer bracket, simplify inside first and apply that outer power last.
- **Step 5:** Rewrite any negative exponent as a positive one (Lesson 3).

**Example 3 — One base at a time**

Simplify the following.

**(a)** $6x^4 \times 3x^5$ **(b)** $(-8a^6)(5a^9)$ **(c)** $\dfrac{24y^{18}}{8y^6}$ **(d)** $\dfrac{40b^{15}}{48b^9}$

**(e)** $(2a^5)(a^6)(4a^2)$ **(f)** $(-20n^7) \div (-4n)$ **(g)** $a^6b^9 \times a^2b^3$ **(h)** $\dfrac{x^7y^5}{x^3y^2}$

**Example 4 — Two or more laws in one chain**

Simplify the following. Each part uses two or more of the exponent laws.

**(a)** $(2x^3)^4$ **(b)** $(-3a^2b^4)^3$ **(c)** $\dfrac{x^4 \cdot x^6}{x^3 \cdot x^2}$

**(d)** $\left(-\dfrac{3a}{b^2}\right)^3$ **(e)** $\left(\dfrac{6a^3 \times 2a^5}{4a^2}\right)^3$ **(f)** $\dfrac{54(a^3b^4)^2}{(3a^2)^3}$

**(g)** $(3a^4b)^2(2a^3b^5)$

**3. Negative Bases: $(-a)^n$ versus $-a^n$**

- $(-a)^n$ means the negative sign is part of the base, so it gets raised to the power too: $(-a)^n = a^n$ when $n$ is even, and $(-a)^n = -a^n$ when $n$ is odd.
- $-a^n$ means only $a$ is the base — the exponent applies to $a$ alone, and the negative sign out front is applied afterward. This is always the opposite sign of $a^n$ itself.

**These are NOT exponent laws — common traps**

| | |
|---|---|
| $(a+b)^2 \neq a^2 + b^2$ | the exponent does not distribute over +: $(2+3)^2 = 25$, but $2^2 + 3^2 = 13$. |
| $2^3 \cdot 3^2 \neq 6^5$ | different bases: the Product Law needs the *same* base. $8 \times 9 = 72$, not 7 776. |
| $a^m + a^n \neq a^{m+n}$ | adding powers is not multiplying them: $2^3 + 2^2 = 12$, but $2^5 = 32$. |

*Coming in Lesson 3:* $4x^{-2} \neq \dfrac{1}{4x^2}$ — an exponent belongs only to the base it touches.

**Example 5 — Negative bases**

Write in simplest form.

**(a)** $(-a)^8 \div (-a)^5$ **(b)** $-a^8 \div (-a)^5$ **(c)** $-a^9 \div (-a)^4$

**(d)** $\dfrac{100x^6y^9(-z^2)^3}{5(2x^2y)^3z^2y^2}$

**4. Extension: Variable Exponents**

The laws still apply when the exponents are algebraic expressions: add or subtract the exponent expressions instead of single numbers.

**Example 6 — Variable exponents**

Simplify each expression.

**(a)** $\dfrac{b^{5x+2y}}{b^{x-3y}}$ **(b)** $\dfrac{x^{4a+9b} \cdot x^{2a+b}}{x^a \cdot x^{3a-4b}}$ **(c)** $\dfrac{m^{2x+5}}{m^6}$

## Lesson 3: Negative Exponents and Scientific Notation

*Extending the exponent laws past $a^0 = 1$ to negative integer exponents — including fractional bases — then using negative powers of 10 to write numbers in scientific notation.*

### 1. The Negative Exponent Pattern

Recall from Lesson 1 that $a^0 = 1$ ($a \neq 0$). Now keep going *past* zero: each time the exponent of 6 drops by 1, the value is divided by 6 once more.

$$6^1 = 6 \qquad 6^0 = 1 \qquad 6^{-1} = \frac{1}{6} \qquad 6^{-2} = \frac{1}{36} \qquad 6^{-3} = \frac{1}{216}$$
$$(6 \div 6) \qquad (1 \div 6) \qquad \left(\frac{1}{6} \div 6\right) \qquad \left(\frac{1}{36} \div 6\right)$$

The same "keep dividing by the base" pattern holds for *any* nonzero base $a$:

$$a^{-1} = \frac{1}{a}, \qquad a^{-2} = \frac{1}{a^2}, \qquad a^{-3} = \frac{1}{a^3}, \qquad \ldots, \qquad a^{-n} = \frac{1}{a^n} \quad (a \neq 0).$$

**Example 1 — Rewriting a single power with a positive exponent**

Write each expression with positive exponents only.

**(a)** $9^{-3}$ **(b)** $x^{-7}$ **(c)** $a^{-m}$

### 2. Where Does the Negative Exponent Law Come From?

The quotient law confirms the pattern. It gives $6^3 \div 6^8 = 6^{3-8} = 6^{-5}$; expanding the same quotient instead gives:

$$6^3 \div 6^8 = \frac{6 \times 6 \times 6}{6 \times 6 \times 6 \times 6 \times 6 \times 6 \times 6 \times 6} = \frac{1}{6^5}$$

(three 6's cancel from the top, leaving five on the bottom). Both describe the same expression, so $6^{-5} = \dfrac{1}{6^5}$. Flipped around, dividing by a negative power is the same as multiplying by the matching positive power: $\dfrac{1}{6^{-5}} = 6^5$.

**Negative Exponent Law:** For any nonzero base $a$ and positive integer $n$,

$$a^{-n} = \frac{1}{a^n} \qquad \text{and} \qquad \frac{1}{a^{-n}} = a^n.$$

**Example 2 — Simplify, write with positive exponents, then evaluate**

Simplify, express with positive exponents, and evaluate each without using a calculator.

**(a)** $5^6 \times 5^{-4}$ **(b)** $2^3 \times 2^{-7}$ **(c)** $\dfrac{1}{3^{-4}}$ **(d)** $\dfrac{8^{-6}}{8^{-8}}$ **(e)** $(5^2)^{-1}$

**Calculator tip (TI-30XIIS)**

- For a negative exponent use the **negative key** [(−)], not the subtract key: `3^(-)5`.
- Put a negative base in brackets: `((-)6)^(-)3`. Without brackets, `(-)6^2` means $-6^2 = -36$ (the calculator does the power before the negative).
- For an exact fraction, press [F◄►D] after the answer: `3^(-)5` [ENTER] [F◄►D] gives $\frac{1}{243}$.

**Example 3 — True or false? Where the negative exponent attaches**

Identify each of the following statements as true or false.

**(a)** $\dfrac{7^5}{7^{-2}} = 7^7$ **(b)** $\dfrac{7^5}{7^{-2}} = 7^3$ **(c)** $x^{-4} = \dfrac{1}{x^4}$ **(d)** $5x^{-2} = \dfrac{1}{5x^2}$

**NOT an exponent law (continued from Lesson 2)**

$4x^{-2} \neq \dfrac{1}{4x^2}$. The exponent $-2$ belongs only to the base it touches, $x$: $4x^{-2} = \dfrac{4}{x^2}$. Only $(4x)^{-2} = \dfrac{1}{16x^2}$ moves the 4 as well.

### 3. Applying the Law to Algebraic Expressions

The law works the same way for a variable base. Apply the product, quotient and power laws first, then rewrite any negative exponent using $a^{-n} = \frac{1}{a^n}$.

**Example 4 — Simplifying algebraic expressions**

Simplify each expression. Write the final answer with positive exponents.

**(a)** $a^{-5} \times a^{-2}$ **(b)** $8x^3 \div 2x^9$ **(c)** $\dfrac{y^4}{3y^{-6}}$

**(d)** $(-3x)^{-3}$ **(e)** $\dfrac{9a^{-4}}{3b^{-2}}$ **(f)** $\dfrac{(2p)^{-2}}{7q^3}$

**Example 5 — A division chain in three variables**

Simplify and write the answer with positive exponents.

$$4x^2y^{-5}z^{-3} \div \frac{12x^7y^4z^{-2}}{x^4y^{-2}z^3}$$

### 4. Negative Exponents with a Fractional Base

When the base is itself a fraction, the negative exponent law gives

$$\left(\frac{3}{7}\right)^{-2} = \frac{1}{\left(\frac{3}{7}\right)^2} = \frac{1}{\frac{9}{49}} = \frac{49}{9} = \left(\frac{7}{3}\right)^2.$$

So raising a fraction to a negative power is the same as *flipping the fraction* and making the exponent positive.

**Fractional Base Law:** For nonzero $a$ and $b$, and positive integer $n$,

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n.$$

**Example 6 — Fractional bases, numerical and variable**

Evaluate (a)–(c) without using a calculator.

**(a)** $\left(\frac{2}{5}\right)^{-3}$ **(b)** $\left(\frac{5}{8}\right)^{-1}$ **(c)** $\left(\frac{7}{2}\right)^{-2}$

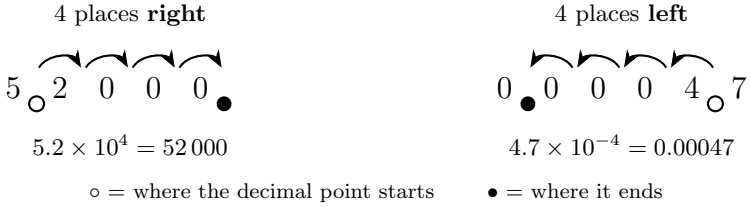
**(d)** Simplify $\left(\frac{2x^2}{3y}\right)^{-3}$. Write the answer with positive exponents.

### 5. Scientific Notation

Negative exponents of 10 are the small place values: $10^{-1} = \frac{1}{10} = 0.1$, $10^{-2} = \frac{1}{100} = 0.01$, $10^{-3} = 0.001$, and so on. A positive number is in **scientific notation** when it is written as $a \times 10^n$, where the **coefficient** $a$ satisfies $1 \le a < 10$ and $n$ is an integer. **Product form** shows where the power of ten comes from:

$$52\,000 = 5.2 \times 10 \times 10 \times 10 \times 10 = 5.2 \times 10^4, \qquad 0.00047 = \frac{4.7}{10 \times 10 \times 10 \times 10} = 4.7 \times 10^{-4}.$$

To change $a \times 10^n$ to standard notation, move the decimal point of $a$ by $n$ places — to the **right** if $n$ is positive, to the **left** if $n$ is negative. A negative exponent gives a small *positive* number, never a negative one.




**Multiplying and dividing.** The exponent laws do the work. To **multiply**, multiply the coefficients and *add* the exponents (product law); to **divide**, divide the coefficients and *subtract* the exponents (quotient law). If the new coefficient is not between 1 and 10, move its decimal point and change the exponent the *other* way by the same number of places: $30 \times 10^7 = 3 \times 10^1 \times 10^7 = 3 \times 10^8$.

**Calculator tip (TI-30XIIS)**

The [EE] key ([2nd] [$x^{-1}$]) means "×10 to the". `4.1` [EE] [(−)] `3` enters $4.1 \times 10^{-3}$. An answer shown as `2.5` $\times 10^{-9}$ means $2.5 \times 10^{-9}$ — never write it as $2.5^{-9}$.

**Example 7 — Scientific notation both ways, then the exponent laws**

Write (a) and (b) in scientific notation and (c) in standard notation. Evaluate (d) and (e), writing each result in scientific notation.

**(a)** 84 200 **(b)** 0.00072 **(c)** $4.1 \times 10^{-3}$

**(d)** $(2.5 \times 10^5)(3 \times 10^4)$ **(e)** $(1.2 \times 10^{-3}) \div (4 \times 10^5)$

## Lesson 4: Rational Exponents and Radicals

*What $a^{1/n}$ and $a^{m/n}$ mean, evaluating them, converting both ways between powers and radicals, and solving problems.*

### 1. What Does $a^{1/n}$ Mean?

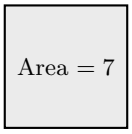
The exponent laws were built for whole-number exponents, but they ought to hold for a fraction too — and that forces a meaning on $a^{1/2}$. Multiply $7^{1/2}$ by itself:

$$7^{1/2} \cdot 7^{1/2} = 7^{\frac{1}{2}+\frac{1}{2}} = 7^1 = 7.$$

By the definition of a square root, $\sqrt{7} \cdot \sqrt{7} = 7$ as well. Both give 7 when multiplied by themselves, so they are the same number: $7^{1/2} = \sqrt{7}$. The same reasoning works for any root — multiply $2^{1/3}$ by itself three times:

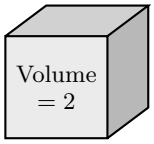
$$2^{1/3} \cdot 2^{1/3} \cdot 2^{1/3} = 2^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}} = 2^1 = 2,$$

which matches $\sqrt[3]{2} \cdot \sqrt[3]{2} \cdot \sqrt[3]{2} = 2$. So $2^{1/3} = \sqrt[3]{2}$.




side $= \sqrt{7} = 7^{1/2}$

*A square of area 7 has side $7^{1/2}$, because $\left(7^{1/2}\right)^2 = 7$.*




edge $= \sqrt[3]{2} = 2^{1/3}$

*A cube of volume 2 has edge $2^{1/3}$, because $\left(2^{1/3}\right)^3 = 2$.*

**The $n$th Root as a Power:** For a natural number $n \geq 2$,

$$a^{1/n} = \sqrt[n]{a}.$$

If $n$ is even, $a$ must be non-negative ($a \geq 0$) for the root to exist in the real numbers; if $n$ is odd, $a$ can be any real number.

### 2. Extending to $a^{m/n}$

The denominator of a rational exponent tells which root to take; the numerator is an ordinary power, applied either before or after the root.

**numerator** $m$: the **power** — **denominator** $n$: the **index** of the root

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$$

base $a$

Both orders give the same answer for $27^{2/3}$.

**Root first, then power**

$$27^{2/3} = \left(27^{1/3}\right)^2 = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$$

**Power first, then root**

$$27^{2/3} = \left(27^2\right)^{1/3} = \sqrt[3]{27^2} = \sqrt[3]{729} = 9$$

Both routes agree, but taking the root first keeps the numbers much smaller ($3^2 = 9$ instead of $\sqrt[3]{729}$), so it is usually the faster path.

**Rational Exponents:** For natural numbers $m$ and $n$ ($n \geq 2$), with $\frac{m}{n}$ in lowest terms,

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}.$$

If $n$ is even, $a$ must be non-negative.

### 3. Negative Rational Exponents

A negative exponent still means "take the reciprocal," $a^{-k} = \dfrac{1}{a^k}$, whether or not the exponent is a fraction: $27^{-2/3} = \dfrac{1}{27^{2/3}} = \dfrac{1}{9}$.

**Negative Rational Exponents:** For natural numbers $m$ and $n$ ($n \geq 2$), with $\dfrac{m}{n}$ in lowest terms,

$$a^{-m/n} = \frac{1}{a^{m/n}} = \frac{1}{\left(\sqrt[n]{a}\right)^m} = \frac{1}{\sqrt[n]{a^m}}, \qquad a \neq 0.$$

If $n$ is even, $a$ must be *positive*.

**When is $a^{m/n}$ defined? (summary)**

Write the exponent $\dfrac{m}{n}$ in **lowest terms** with $n \geq 2$. Then $a^{m/n}$ is a real number

- when $n$ is **even**: for $a \geq 0$ ($a > 0$ if the exponent is negative);
- when $n$ is **odd**: for any real $a$ ($a \neq 0$ if the exponent is negative).

Reduce first: $(-8)^{2/6}$ must be treated as $(-8)^{1/3} = -2$. (Squaring first, $\sqrt[6]{(-8)^2} = \sqrt[6]{64} = 2$, gives the wrong sign.)

### 4. Evaluating Rational-Exponent Powers

**Example 1 — Evaluating numerical powers without a calculator**

Evaluate each power without using a calculator.

**(a)** $16^{3/2}$ **(b)** $1\,000^{2/3}$ **(c)** $8^{-4/3}$ **(d)** $\left(5^2 + 12^2\right)^{1/2}$

**Example 2 — Negative bases and the placement of the minus sign**

**(a)** Evaluate $(-27)^{2/3}$.

**(b)** Evaluate $-\left(27^{2/3}\right)$.

**(c)** Exactly one of the following powers has no meaning (as a real number). Identify which one, and explain why the other three can be evaluated.

A. $27^{2/3}$ B. $(-27)^{2/3}$ C. $16^{3/2}$ D. $(-16)^{3/2}$

**Example 3 — Fractional bases**

Evaluate each power. Express your answer as a fraction in lowest terms.

**(a)** $\left(\dfrac{16}{25}\right)^{3/2}$ **(b)** $\left(\dfrac{16}{25}\right)^{-3/2}$

**Example 4 — A cube of known volume**

A cube has a volume of 100 $cm^3$.

**(a)** Write a power that represents the edge length of the cube.

**(b)** Write a power that represents the surface area of the cube.

**(c)** Use a calculator to find the edge length and surface area, each to the nearest tenth.

**Calculator tip (TI-30XIIS)**

- Put a fractional exponent in brackets: `8^(2÷3)` gives 4. Typing `8^2÷3` means $8^2 \div 3 = \frac{64}{3}$.
- Put a negative base in brackets and use the negative key [(−)], not subtract: `((-)8)^(1÷3)` gives $-2$.
- A negative base only works when the exponent is 1 over an odd number (like $\frac{1}{3}$); `((-)8)^(2÷3)` gives a `DOMAIN` error. Take the root first, then the power: `(((-)8)^(1÷3))` [$x^2$] gives 4.
- For an exact fraction press [F◄►D] after the answer: `27^((-)2÷3)` [ENTER] [F◄►D] gives $\frac{1}{9}$.
- An $n$th root can also be typed with [2nd] [^] ($\sqrt[x]{\ }$), index first: `3` [2nd] [^] `64` gives 4.

**Example 5 — Forcing a chosen exponent, and converting to radicals**

**(a)** Write 9 as a power with an exponent of $\frac{1}{2}$, and again as a power with an exponent of $\frac{1}{3}$.

**(b)** Write each expression as an equivalent radical.

**(i)** $x^{1/5}$ **(ii)** $y^{3/8}$ **(iii)** $z^{-1/4}$ **(iv)** $w^{-5/6}$

## 5. Writing Powers as Radicals

**Example 6 — Where the minus sign and the coefficient belong**

Write each power as a radical.

**(a)** $m^{1/5}$ **(b)** $-n^{3/7}$ **(c)** $(-p)^{5/3}$

**(d)** $(-p)^{-5/3}$ **(e)** $4s^{2/3}$ **(f)** $(4s)^{2/3}$

## 6. Writing Radicals as Powers

The same definition runs in reverse: any radical can be rewritten as a power with a rational exponent by matching the index of the root to the denominator of the exponent, $\sqrt[n]{x^m} = x^{m/n}$.

**Example 7 — From radical form back to a rational exponent**

Write each radical as a power in the form $k^n$, where $n \in Q$.

**(a)** $\sqrt[4]{k^3}$ **(b)** $\sqrt[6]{k^5}$ **(c)** $\sqrt{k^7}$ **(d)** $\dfrac{1}{\sqrt[3]{k^5}}$

## Lesson 5: Simplifying with Rational Exponents

*The exponent laws with fractional exponents — single laws, roots as single powers, and multi-law chains.*

In Lesson 4, $a^{m/n} = (\sqrt[n]{a})^m$ turned rational exponents into radicals and back. Now the exponent laws from Lessons 1–3 do the simplifying: they work the same way with fractional exponents — you just add, subtract or multiply fractions.

### 1. Reviewing the Exponent Laws

Simplifying algebraic expressions with rational exponents just means reusing the laws already established for integer exponents. They hold for any rational exponents when the base is positive; with a negative base, take the root first.

| | | | |
|---|---|---|---|
| Product Law | $x^m \cdot x^n = x^{m+n}$ | Power of a Product Law | $(xy)^m = x^m y^m$ |
| Quotient Law | $x^m \div x^n = x^{m-n}$ | Power of a Quotient Law | $\left(\dfrac{x}{y}\right)^m = \dfrac{x^m}{y^m},\ y \neq 0$ |
| Power of a Power Law | $(x^m)^n = x^{mn}$ | Negative Exponent Law | $x^{-m} = \dfrac{1}{x^m},\ x \neq 0$ |

### 2. Simplifying Expressions with Rational Exponents

The laws from Section 1 apply directly — there is no need to convert to radical form first. Convert to a radical only at the very end, once the exponent has been fully simplified.

**Example 1 — The exponent laws with fractional exponents**

Simplify each expression. Write your answer as a power with a positive exponent, and then in radical form.

**(a)** $w^{5/4} \times w$ **(b)** $t^{1/4} \div t^{7/4}$ **(c)** $\left(b^{1/3}\right)^{3/5}$ **(d)** $\left(\dfrac{r^2}{s}\right)^{-1/3}$

**(e)** $6c^{5/6} \times 2c^{-1/3}$ **(f)** $\dfrac{7d^{4/5}}{49d^{-3/5}}$ **(g)** $\left(27e^{1/2}\right)^{4/3}$

### 3. Roots as Single Powers

Any root can be written as a single power. A "root of a root" multiplies the indices together (a square root of a cube root is a sixth root, and so on), and a root in a denominator becomes a negative exponent.

**Example 2 — One power, then one evaluation**

Write each expression as a single power, then evaluate.

**(a)** $\sqrt{\sqrt{2401}}$ **(b)** $\dfrac{1}{\sqrt{361}}$ **(c)** $\sqrt[3]{\sqrt{729}}$

### 4. Writing Radicals in the Form $ax^n$

When a radical contains both a perfect-power numerical factor and a variable factor, pull the numerical part out as an integer coefficient $a$, leaving the variable part as a power $x^n$.

**Example 3 — Splitting off the numerical factor**

Write each expression in the form $ax^n$, where $a \in I$ and $n \in Q$.

**(a)** $\sqrt[3]{64x^8}$ **(b)** $\sqrt[4]{16x^6}$ **(c)** $\sqrt{400x}$

**(d)** $\left(\sqrt[3]{x^7}\right)\left(\sqrt[3]{x^2}\right)$ **(e)** $3\sqrt{x} \times 2\sqrt[3]{x}$

### 5. Simplifying Nested Radicals and Powers

**Example 4 — Nested roots carrying variables**

Write each expression using rational exponents (in the form $ax^n$).

**(a)** $\sqrt{\sqrt{k^5}}$ **(b)** $\sqrt{\sqrt[3]{729w^9}}$ **(c)** $\left(\sqrt[4]{k^6y^{10}}\right)^{3/2}$

### 6. Multi-Law Chains with Rational and Negative Exponents

Use the same order as in Lesson 2: raise each bracket to its power (every factor, coefficient included), combine coefficients, add or subtract exponents base by base, and finish with positive exponents. Evaluate a numerical coefficient such as $8^{2/3}$ by taking the root first.

**Example 5 — Rational and negative exponents together**

Simplify. Write each answer with positive exponents.

**(a)** $\left(8x^6y^{-3}\right)^{2/3} \cdot x^{-1/2}$ **(b)** $\dfrac{\left(81a^{-8}b^2\right)^{1/4}}{3a^{-1/2}}$

## Lesson 6: Exponents Unit Review

*Every law from the unit in one place — vocabulary, the product and quotient laws, zero and negative exponents, scientific notation, and rational exponents.*

## 1. Laws at a Glance

| | | | |
|---|---|---|---|
| **Product Law** | $a^m \times a^n = a^{m+n}$ | **Power of a Power Law** | $(a^m)^n = a^{mn}$ |
| **Quotient Law** | $a^m \div a^n = a^{m-n} \quad (a \neq 0)$ | **Power of a Product Law** | $(ab)^n = a^n b^n$ |
| **Zero Exponent Law** | $a^0 = 1 \quad (a \neq 0)$ | **Power of a Quotient Law** | $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \quad (b \neq 0)$ |
| **Negative Exponent Law** | $a^{-n} = \frac{1}{a^n} \quad (a \neq 0)$ | **Fractional Base Law** | $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$ |
| **Rational Exponent** | $a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$ | **Negative Rational Exponent** | $a^{-m/n} = \frac{1}{a^{m/n}} \quad (a \neq 0)$ |

*Key reminders:* the Product and Quotient Laws combine exponents only when the bases are **multiplied** or **divided**, never for terms added or subtracted: $b^6 + b^2$ is not a single power. For $a^{m/n}$, write $\frac{m}{n}$ in lowest terms; if $n$ is even, $a$ must not be negative.

## 2. Vocabulary: Base, Exponent, and Coefficient

In a power $a^n$, the **base** $a$ is the factor being repeated and the **exponent** $n$ tells how many times it is used as a factor; in a term like $-\frac{4y^6}{9}$ the **coefficient** is the number multiplying the power, here $-\frac{4}{9}$. Negative signs need care: in $-a^0$ the exponent applies only to $a$, so $-a^0 = -(a^0) = -1$ (for $a \neq 0$), while in $(-a)^0 = 1$ the entire quantity $-a$ is the base.

**Example 1 — Base, exponent, coefficient, and the $-a^0$ trap**

**(a)** State the base and exponent in the power $(-3)^5$.

**(b)** State the coefficient in $-\frac{4y^6}{9}$.

**(c)** Evaluate $-a^0$ and $(-a)^0$, assuming $a \neq 0$.

## 3. Expanding Powers as Repeated Multiplication

Expanding a power as repeated multiplication is a good check on a simplification. A coefficient sitting *outside* a bracket is not raised to the power; only what is inside is.

**Example 2 — True or false: two expansions**

Consider the following two statements.

Statement 1: $8p^3q^5 = 8 \times p \times p \times p \times q \times q \times q \times q \times q$ Statement 2: $4(mn)^3 = 4 \times 4 \times 4 \times m \times m \times m \times n \times n \times n$

Determine whether each statement is true or false, and explain your reasoning.

## 4. Which Law Applies? A Concept Check

**Example 3 — Sorting valid simplifications from invalid ones**

Which of the following expressions simplify to $b^8$?

I. $b^5 \times b^3$ II. $b^{10} - b^2$ III. $b^{10} \div b^2$ IV. $b^4 + b^4$

### 5. Product and Quotient Laws with Coefficients

With a coefficient attached, multiply or divide the coefficients separately from adding or subtracting the exponents.

**Example 4 — The Product Law with coefficients**

Simplify $6b^4 \times 3b^5$.

**Example 5 — The Quotient Law, including a negative exponent**

Simplify each quotient.

**(a)** $\dfrac{8x^{14}}{4x^6}$ **(b)** $\dfrac{9y^2}{3y^{-5}}$

### 6. Negative Exponents

A negative exponent tells you to take the reciprocal of the base raised to the corresponding positive exponent — it does *not* make the value negative, and it does not affect a coefficient that sits outside the power.

**Example 6 — Rewriting with positive exponents**

Rewrite each expression using only positive exponents.

**(a)** $y^{-4}$ **(b)** $7x^{-3}$

### 7. Rational Exponents

A rational (fractional) exponent combines a root and a power: $a^{m/n} = \sqrt[n]{a^m}$, where the denominator $n$ becomes the index of the root and the numerator $m$ becomes the power. A *negative* rational exponent still means "reciprocal," exactly as with integer exponents.

**Example 7 — Radical form, and evaluating a rational exponent**

**(a)** Express $x^{2/7}$ in radical form. **(b)** If $p = 9$ and $q = -27$, evaluate $p^{3/2} - q^{-2/3}$ to the nearest hundredth.

**Example 8 — Simplifying with a rational exponent**

Simplify $\left(16x^8y^{-4}\right)^{3/4}$. Write the answer with positive exponents.

### 8. Scientific Notation

A number in scientific notation has the form $a \times 10^n$ with $1 \le a < 10$. To multiply two such numbers, multiply the coefficients and add the exponents of 10, then check the result still satisfies $1 \le a < 10$, adjusting the exponent if it does not.

**Example 9 — Multiplying in scientific notation**

Evaluate $(5.4 \times 10^{-2}) \times (3.0 \times 10^{6})$, and write the answer in proper scientific notation.

**Example 10 — A stack of thin sheets**

A sheet of aluminum foil is $6.4 \times 10^{-5}$ m thick. Approximately how many sheets are needed to build a stack 1.6 m high?